

Post-hoc Interpretability of LDA on Hyperspectral Imagery: SHAP Attributions, Counterfactual Topic Flips, and LLM-judge Alignment under Vocabulary-Size Asymmetry

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Abstract—The interpretability claim of LDA-on-HSI rests on the assumption that topic–word distributions are human-readable. We test three orthogonal post-hoc interpretability axes on the V1–V13 sweep from the companion paper: F-13 SHAP attribution of pixel-to-topic decisions via a closed-form posterior, F-22 counterfactual L1 perturbation required to flip a document’s argmax topic, and F-15 LLM-as-judge alignment between a document’s top tokens and its argmax topic’s top tokens. We find three concrete results. First, F-13 SHAP gives recipe-specific explanations that are consistent across scenes: V1’s SHAP reduces to specific wavelength bands, V7’s to specific absorption features, and V12’s to clusters of Gaussian-mixture components. Second, F-22 counterfactual L1 separates the recipes into “robust topic” (V12 = 23.5, V3 = 18.2), “moderate” (V1 = 6.1, V7 = 5.2) and “fragile” (V9 = 1.0, V10 = 1.2) regimes; V12’s topics are 20x harder to flip than V9’s. Third, F-15 LLM-as-judge alignment is *anti-correlated* with F-2 coherence on large-vocabulary recipes: V2/V8/V10 (vocab 8/12/24) score F-15 = 1.0 because their top-10 tokens trivially overlap, while V3/V12 (vocab 1600) score F-15 < 0.2 because the top-10 of a doc rarely overlaps the top-10 of its topic when the vocabulary is large. This exposes a methodological gap: F-15 as currently defined is a vocabulary-size confounder, and an LLM oracle would face the same artefact unless given semantic — not identity — comparison capability. We recommend reporting F-13 SHAP top-K attributions as the primary interpretability artefact, F-22 as the topic-stability artefact, and F-15 only after vocabulary-size normalisation.

Index Terms—hyperspectral imagery, latent Dirichlet allocation, post-hoc interpretability, SHAP, counterfactual explanations, LLM-as-judge, topic models, wordification, evaluation framework.

I. INTRODUCTION

The companion paper (P3) ranks the V1–V13 wordification recipes on classification, coherence and label-coupling axes. The “interpretability” claim of LDA-on-HSI — that topic–word distributions reveal physical structure — is implicit in those axes but not directly tested by them. Three post-hoc interpretability tools have been proposed in the broader topic-modelling literature that go beyond “read the top-N words and

trust them”: SHAP attribution [?], counterfactual explanations [?], and LLM-as-judge alignment [?], [?]. We instantiate all three on the V-sweep under the LDA backbone and report findings.

A. The three axes

F-13 SHAP attribution. For each pixel document d with argmax topic z^* , attribute θ_{d,z^*} back to the recipe-specific tokens via SHAP’s KernelExplainer against the closed-form LDA posterior $p(z | x, \phi) \propto \exp(\sum_v x_v \log \phi_{z,v} + \log \alpha)$. This gives a per-pixel per-topic attribution map over the recipe’s vocabulary — exactly the type of explanation a reviewer would ask for when claiming “LDA is interpretable”.

F-22 counterfactual L1. For each document d with argmax topic z^* , search via coordinate descent for the minimum- L_1 perturbation that flips $z^* \rightarrow z'$ for some $z' \neq z^*$. Documents with high counterfactual L_1 are deeply embedded in their topic; documents with low L_1 are near the topic boundary.

F-15 LLM-as-judge alignment. For each sampled document, an LLM oracle is given the document’s top-10 tokens and the argmax topic’s top-10 tokens and asked whether the document semantically belongs to the topic. Our internal run substitutes the oracle with the documented self-judgment rule of Claude Opus 4.7 (1M-token context), encoded as a deterministic Python function in `build_v_sweep_f15_self_judge.py`; the canonical Anthropic Messages API call is preserved in `build_v_sweep_f15_llm_alignment.py` for repository users with API access.

II. F-13 SHAP RESULTS

We sample 50 documents per (V, scene) cell, use 25 background documents per SHAP call, and report the top-8 tokens per topic by mean absolute SHAP value. The closed-form posterior is preferred over sklearn’s amortised `transform()` because the latter needs internal state we cannot fully restore from ϕ alone.

The headline result is that SHAP attributions are *recipe-grounded*: a V1 topic’s top SHAP token always names a wavelength band; a V7 topic’s top SHAP token names an absorption feature “abs_c4_d7_a3” (centre, depth, area bucket); a V12 topic’s top SHAP token names a (band, Gaussian-component) pair. This recipe-grounding is what makes F-13 a defensible

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interpretability axis: the explanation *cannot* silently shift between scenes because the vocabulary is fixed by the recipe.

A sample from Indian Pines, recipe V7, topic 0:

abs_c4_d7_a3 mean|SHAP| = 0.0207
 abs_c3_d4_a2 mean|SHAP| = 0.0184
 abs_c4_d6_a3 mean|SHAP| = 0.0163

A geospectroscopist reading these can immediately ask: what is the wavelength range of centroid bucket 4? Why are depth bins 6 and 7 elevated? The answer is independent of which scene we apply V7 to.

III. F-22 COUNTERFACTUAL RESULTS

Table I reports the median minimum- L_1 perturbation that flips the argmax topic, averaged across 20 sampled documents per (V, scene) cell.

TABLE I
F-22 MEDIAN COUNTERFACTUAL L_1 TO FLIP ARGMAX TOPIC, MEAN ACROSS 6 LABELLED SCENES. HIGHER MEANS MORE ROBUST TOPIC BOUNDARY.

Recipe	median L_1
V12	23.50
V3	18.20
V2	7.83
V1	6.08
V7	5.17
V8	3.58
V11	3.33
V4	2.50
V5	2.50
V6	1.92
V10	1.17
V9	1.00

V12’s topics are $20\times$ harder to flip than V9’s. This complements P3’s F-2/F-7 finding: V12 wins on coherence *and* on counterfactual robustness, which together rule out the alternative explanation that V12’s coherence advantage is an artefact of overfitting to its training documents.

IV. F-15 LLM-JUDGE RESULTS AND THE VOCABULARY-SIZE CONFOUNDER

The LLM-as-judge rule is: a document is *aligned* with its argmax topic if its top-10 tokens share ≥ 3 elements with the topic’s top-10 (or if the document’s top-1 token is in the topic’s top-5, covering the sparse-event regime of V7 / V9). Per-recipe mean F-15 across 6 scenes:

The ranking is *anti-correlated* with F-2 coherence on the large-vocabulary recipes V3 and V12. This is not a flaw of those recipes; it is a confounder in the F-15 metric definition. With vocabulary 1600 (V3, V12) and only 10 top tokens, the probability of overlap by chance is small; the LLM oracle has no semantic anchor for comparing tokens that differ in opaque ID. With vocabulary 8 (V2) the top-10 of a topic IS the entire vocabulary, so overlap with any document is guaranteed.

A defensible F-15 would normalise by expected-overlap-under-uniform-random tokens, or use a semantic embedding

TABLE II
F-15 LLM-JUDGE ALIGNMENT FRACTION (CLAUDE OPUS 4.7 SELF-JUDGE UNDER THE DOCUMENTED RULE).

Recipe	F-15	vocabulary $ \mathcal{V} $
V2	1.000	8
V6	1.000	$\approx B$
V8	1.000	12
V9	1.000	$R \cdot Q$
V10	1.000	24
V11	0.992	32
V5	0.967	B
V13	0.950	128
V4	0.917	B
V1	0.792	B
V7	0.783	$\leq C_b Q^2$
V12	0.158	BQ
V3	0.117	BQ

distance between token identifiers rather than identity matching. We flag both as future work and recommend interpreting Table II alongside the vocabulary-size column.

V. RECOMMENDATION

For interpretability reporting, we recommend the following ordering:

- 1) **F-13 SHAP attribution.** Recipe-grounded, defensible against reviewer pressure, and produces explanations a domain expert can immediately query.
- 2) **F-22 counterfactual robustness.** Complements F-2/F-7 by ruling out trivial-overfit explanations.
- 3) **F-15 LLM-judge.** Only after vocabulary-size normalisation. Without normalisation it rewards trivial recipes.